```
import java.util.Scanner;

public class Car {

    private String brand;
    private String model;
    private int year;
    private String color;
    private double mileage;
    private double fuelLevel;
    private boolean engineOn;

    // Default constructor
    public Car() {
        brand = "Unknown";
        model = "Unknown";
        year = 0;
        color = "Unknown";
        mileage = 0;
        fuelLevel = 0;
        engineOn = false;
    }

    // Parameterized constructor
    public Car(String brand, String model, int year, String color,
            double mileage, double fuelLevel, boolean engineOn) {

        this.brand = brand;
        this.model = model;
        this.year = year;
        this.color = color;
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    this.mileage = mileage;
    this.fuelLevel = fuelLevel;
    this.engineOn = engineOn;
}

// Getters
public String getBrand() {
    return brand;
}

public String getModel() {
    return model;
}

public int getYear() {
    return year;
}

public String getColor() {
    return color;
}

public double getMileage() {
    return mileage;
}

public double getFuelLevel() {
    return fuelLevel;
}

public boolean isEngineOn() {
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    return engineOn;
}

// Setters
public void setBrand(String brand) {
    this.brand = brand;
}

public void setModel(String model) {
    this.model = model;
}

public void setYear(int year) {
    this.year = year;
}

public void setColor(String color) {
    this.color = color;
}

public void setMileage(double mileage) {
    this.mileage = mileage;
}

public void setFuelLevel(double fuelLevel) {
    this.fuelLevel = fuelLevel;
}

public void setEngineOn(boolean engineOn) {
    this.engineOn = engineOn;
}
```

```
// Start engine
public void startEngine() {
    if (!engineOn) {
        engineOn = true;
        System.out.println("The engine has started.");
    } else {
        System.out.println("The engine is already running.");
    }
}

// Stop engine
public void stopEngine() {
    if (engineOn) {
        engineOn = false;
        System.out.println("The engine has stopped.");
    } else {
        System.out.println("The engine is already off.");
    }
}

// Drive the car
public void drive(double distance) {
    if (distance <= 0) {
        System.out.println("Distance must be more than 0.");
    } else if (!engineOn) {
        System.out.println("Start the engine first.");
    } else if (fuelLevel <= 0) {
        System.out.println("No fuel left.");
    } else {
        double fuelNeeded = distance * 0.08;
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        if (fuelNeeded > fuelLevel) {
            double possibleDistance = fuelLevel / 0.08;

            mileage = mileage + possibleDistance;
            fuelLevel = 0;

            System.out.println("The car ran out of fuel.");
        } else {
            mileage = mileage + distance;
            fuelLevel = fuelLevel - fuelNeeded;

            System.out.println("The car drove " + distance + " km.");
        }
    }
}

// Refuel the car
public void refuel(double amount) {
    if (amount <= 0) {
        System.out.println("Fuel amount must be more than 0.");
    } else {
        fuelLevel = fuelLevel + amount;

        if (fuelLevel > 100) {
            fuelLevel = 100;
        }

        System.out.println("Fuel level: " + fuelLevel + "%");
    }
}
```

```
// Show car information
public void displayInfo() {
    System.out.println("\n--- Car Information ---");
    System.out.println("Brand: " + brand);
    System.out.println("Model: " + model);
    System.out.println("Year: " + year);
    System.out.println("Color: " + color);
    System.out.println("Mileage: " + mileage + " km");
    System.out.println("Fuel Level: " + fuelLevel + "%");
    System.out.println("Engine On: " + engineOn);
}

public static void main(String[] args) {

    Scanner scanner = new Scanner(System.in);

    System.out.println("Choose a car:");
    System.out.println("1. Proton Saga");
    System.out.println("2. Perodua Myvi");
    System.out.print("Enter choice: ");

    int choice = scanner.nextInt();

    Car car;

    if (choice == 1) {
        car = new Car(
            "Proton",
            "Saga",
            2022,
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        "White",
        0,
        100,
        false
    );

} else if (choice == 2) {
    car = new Car(
        "Perodua",
        "Myvi",
        2022,
        "Red",
        0,
        100,
        false
    );

} else {
    System.out.println("Invalid choice.");
    scanner.close();
    return;
}

car.displayInfo();

System.out.println(
    "\nSelected car: " + car.getBrand() + " " + car.getModel()
);

// Setter example
car.setColor("Black");
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        System.out.println("New color: " + car.getColor());

        System.out.print("\nStart engine? 1 = Yes, 0 = No: ");
        int start = scanner.nextInt();

        if (start == 1) {

            car.startEngine();

            for (int i = 1; i <= 3; i++) {
                System.out.print("Enter distance to drive: ");
                double distance = scanner.nextDouble();

                car.drive(distance);
            }

            System.out.print("Enter fuel amount to add: ");
            double fuel = scanner.nextDouble();

            car.refuel(fuel);

        } else {
            System.out.println("Engine remains off.");
        }

        car.displayInfo();
        car.stopEngine();

        scanner.close();
    }
```

}